25901642 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCCSC100 - DATA VISUALIZATION USING PYTHON

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Find the function used to create a basic line plot in Matplotlib? [K] / [CO1]
2. What is Seaborn, and why is it used for data visualization? [K] / [CO1]
3. Name two types of input components available in Dash. [K] / [CO2]
4. Name any 2 static plotting library. [K] / [CO2]
5. Can we save the charts as image files in Plotly? Describe how. [U] / [CO2]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. Relate data visualization techniques to real-world decision-making processes. [U] / [CO1]

7. Discuss different methods used in Plotly express for quick visualization with necessary codes. [U] / [CO2]
8. What is a multi-plot grid in Seaborn? How is it useful? [K] / [CO1]
9. Describe the role of storytelling in data visualization and how it improves comprehension. [U] / [CO2]
10. Explain the different types of charts and their appropriate use cases in data visualization. Use appropriate diagrams. [U] / [CO2]
11. Compare the effectiveness of line plots and scatter plots in data representation. [U] / [CO1]
12. Differentiate Plotly Express and Plotly Grpah Objects. [U] / [CO2]

[5x5 = 25]